[Aim: 100/100 in Maths]

V. Imp.

$$\boxed{a_n = a + (n-1)d}$$

आरंभ

ARITHMETIC PROGRESSION

Lecture #5

MT

① Position of MT.

$n \rightarrow$ odd $\rightarrow$ Pos $\Rightarrow \left(\frac{n+1}{2}\right)^{th}$

$n \rightarrow$ even $\rightarrow$ Pos $\Rightarrow \left(\frac{n}{2}\right)^{th}$ & $\left(\frac{n}{2}+1\right)^{th}$

② Value of MT.

#LP: How many numbers of two digit are divisible by 7 ?

Two digits : 10 ———— 99

div by 7 : $14, 21, 28 - - - - - - 98$ ← a_n → AP

$a_1 \quad a_2 \quad a_3$

$$7\overline{)99}\,(14$$
$$\frac{7}{29}$$
$$\frac{28}{1}$$ ← Remainder

$99 - 1 \Rightarrow 98$

$$7\overline{)98}\,(14$$
$$\frac{7}{28}$$
$$\frac{28}{0}$$

$d_1 = a_2 - a_1 = 21 - 14 = 7$

$d_2 = a_3 - a_2 = 28 - 21 = 7$

c.d → same

It's an AP

Let n terms are there

$\therefore a_n = 98$

$a + (n-1)d = 98$

$14 + (n-1)(7) = 98$

$(n-1)(7) = 98 - 14$

$(n-1)(\cancel{7}) = \cancel{84}\,12$

$n - 1 = 12$

$n = 13$

$\therefore$ 13 two digit numbers are div by 7.

$$7\overline{)99}\,(14$$
$$\frac{7}{29}$$
$$\frac{28}{2}$$

#LP: Find the number of integers between 50 and 500 which are divisible by 7.

50 ——— 500

div. by 7 :- 56, 63, 70 - - - - - 497

AP

a_n

n

Let there are n term in this AP.

$\therefore a_n = 497$

$a+(n-1)d = 497$

$56 + (n-1)(7) = 497$

$(n-1)(7) = 497 - 56$

$(n-1)(\cancel{7}) = \cancel{441}\ 63$

$\boxed{n = 64}$

$\therefore$ there are 64 integ b/w 50 and 500 which are div by 7

$7\overline{)500}$ 71

49

10

7

3 ← Rem

$500 - 3 \Rightarrow 497$

div by 7

#LP: Two A.P have the same common difference. The first term of one of these is 3, and that of the other is 8. What is the difference between their

i. 2^{nd} terms ? ii. 4^{th} terms ? iii. 10^{th} terms ? iv. 30^{th} terms ?

AP_1	AP_2
d	d
$a = 3$	$a' = 8$

(i) $a_2 - a'_2$

$\Rightarrow [a+(2-1)d] - [a'+(2-1)d]$

$\Rightarrow a + \cancel{d} - a' - \cancel{d}$

$\Rightarrow a - a' \Rightarrow 3 - 8 \Rightarrow -5$

(ii) $a_4 - a'_4$

$[a+3d] - [a'+3d]$

$a + \cancel{3d} - a' - \cancel{3d}$

$a - a' \Rightarrow -5$

(iii) $a_{10} - a'_{10}$

$(a+9d) - (a'+9d)$

$a + \cancel{9d} - a' - \cancel{9d}$

$\Rightarrow a - a' \Rightarrow -5$

(iv) $a_{30} - a'_{30}$

$\Rightarrow -5$

If two APs have same C.d.(d) then diff b/w their any two same terms will be same

#LP: Two A.P have the same common difference. The ~~common~~ difference between their 100^{th} terms is 111 222 333. What is the difference between their Millionth terms ?

A. 0

B. 111 222 333

C. 333 222 111

D. None of the above

#LP: If the p^{th} term of an A.P. is q and the q^{th} term is p, prove that its nth term is (p + q - n) [CBSE 2008 , 2017]

$$a_p = q \Rightarrow a + (p-1)d = q \Rightarrow a + pd - d = q \quad \text{— (I)}$$

$$a_q = p \Rightarrow a + (q-1)d = p \Rightarrow a + qd - d = p \quad \text{— (II)}$$

$$pd - qd = q - p$$

$$d(p-q) = q - p$$

$$d(\cancel{p-q}) = -(\cancel{p-q})1$$

$$d = -1$$

put in (I)

$$a + p(-1) - (-1) = q$$

$$a - p + 1 = q \Rightarrow \boxed{a = p + q - 1}$$

To find $\Rightarrow$ $a_n = a + (n-1)d$

$$= (p+q-1) + (n-1)(-1)$$

$$= p + q \cancel{-1} - n \cancel{+1}$$

$$\boxed{a_n = p + q - n}$$

(H.P.)

#LP: If the m^{th} term of an A.P. be 1/n and n^{th} term be 1/m, then show that its $(mn)^{th}$ term is 1.

$$a_m = \frac{1}{n} \Rightarrow a+(m-1)d = \frac{1}{n} \Rightarrow \cancel{a} + md - \cancel{d} = \frac{1}{n} \quad \text{—(1)}$$

$$a_n = \frac{1}{m} \Rightarrow a+(n-1)d = \frac{1}{m} \Rightarrow \cancel{a} + nd - \cancel{d} = \frac{1}{m} \quad \text{—(2)}$$

$$md - nd = \frac{1}{n} - \frac{1}{m}$$

$$d(\cancel{m-n}) = \frac{\cancel{m-n}}{nm}$$

$$\boxed{d = \frac{1}{mn}}$$

put in (2)

$$a + \cancel{n}\left(\frac{1}{m\cancel{n}}\right) - \frac{1}{mn} = \frac{1}{n}$$

$$a + \cancel{\frac{1}{n}} - \frac{1}{mn} = \cancel{\frac{1}{n}}$$

$$\boxed{a = \frac{1}{mn}}$$

To find :-

$$a_{mn} = a + (mn-1)d$$

$$= \frac{1}{mn} + (mn-1)\frac{1}{mn}$$

$$= \cancel{\frac{1}{mn}} + \cancel{mn} \times \frac{1}{\cancel{mn}} - \cancel{\frac{1}{mn}}$$

$$\boxed{a_{mn} = 1}$$

HP

#LP: If m times the m^{th} term of an A.P. is equal to n times its n^{th} term, show that the $(m+n)^{th}$ term of the A.P. is zero. [CBSE 2008]

$\Rightarrow m \times a_m = n \times a_n$

$\Rightarrow m[a+(m-1)d] = n[a+(n-1)d]$

$\Rightarrow m[a+md-d] = n[a+nd-d]$

$\Rightarrow ma+m^2d-md = na+n^2d-nd$

$\Rightarrow ma+m^2d-md-na-n^2d+nd=0$

$\Rightarrow ma-na+m^2d-n^2d-md+nd=0$

$\Rightarrow a(m-n)+d[m^2-n^2-m+n]=0$

$\Rightarrow a(m-n)+d[(m+n)(m-n)-(m-n)]=0$

$\Rightarrow a(m-n)+d(m-n)[m+n-1]=0$

$(m-n)[a+d(m+n-1)]=0$

$m-n=0$

$m=n$

$\times$

$\boxed{a+(m+n-1)d=0}$

To prove

$a_{(m+n)}=0$

$\boxed{a+(m+n-1)d=0}$

To prove

#LP: If pth , qth and rth terms of an A.P. are a , b , c respectively , then show that: **[CBSE 2016]**

i. $a(q-r)+b(r-p)+c(p-q)=0$

ii. $(a-b)r+(b-c)p+(c-a)q=0$

(ii) → HW

$a_p = a$ | $a_q = b$ | $a_r = c$

$\boxed{[A+(p-1)D] = a}$ | $\boxed{A+(q-1)D = b}$ | $\boxed{A+(r-1)D = c}$

(i) $a(q-r)+b(r-p)+c(p-q)=0$

LHS $[A+(p-1)D](q-r)+[A+(q-1)D](r-p)+[A+(r-1)D](p-q)$

$\Rightarrow [A+pD-D](q-r)+[A+qD-D](r-p)+[A+rD-D](p-q)$

$\Rightarrow \cancel{Aq}+\cancel{pqD}-\cancel{qD}-\cancel{rA}-\cancel{rpD}+\cancel{rD}+\cancel{rA}+\cancel{rqD}-\cancel{rD}-\cancel{pA}-\cancel{pqD}+\cancel{pD}+\cancel{pA}+\cancel{prD}-\cancel{pD}-\cancel{qA}-\cancel{qrD}+\cancel{qD}$

$=0=$ RHS HP

THANK YOU COODIESS !!